

# EXPERIMENT 15: PERFORM TRANSFORMATION USING DIRECT LINEAR TRANSFORMATION (DLT)

## Aim

To perform image transformation using the Direct Linear Transformation (DLT) algorithm in Python and OpenCV.

## Algorithm

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Select four corresponding source and destination points.
4. Compute the homography matrix using `cv2.findHomography()` (DLT method).
5. Apply `cv2.warpPerspective()` to transform the image.
6. Display and save the transformed image.

## Python Code

```
import cv2
import numpy as np

# Read the image
img = cv2.imread("input.jpg")

if img is None:
    print("Error: Image not found")
    exit()

height, width = img.shape[:2]

# Source points
src_pts = np.float32([
    [30, 30],
    [width-30, 30],
    [width-30, height-30],
    [30, height-30]
])
```

```
# Destination points
dst_pts = np.float32([
    [80, 50],
    [width-80, 20],
    [width-20, height-40],
    [50, height-10]
])

# Compute Homography using Direct Linear Transformation
H, mask = cv2.findHomography(src_pts, dst_pts, method=0)

# Apply transformation
output = cv2.warpPerspective(img, H, (width, height))

cv2.imwrite("dlt_output.jpg", output)

cv2.imshow("Original Image", img)
cv2.imshow("DLT Output", output)

print("Homography Matrix:")
print(H)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

## **Input**



## Output



## Result

Thus, the Direct Linear Transformation (DLT) was successfully performed and the transformed image was obtained.